

DSCI 551 Quiz 1

X is a random variable; x is a possible/realized value. Random variables may be categorical, though moments require numerical values.

Core notation and probability rules

| Item                          | Rule / meaning                                                                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Sample space                  | S: all outcomes; P(S) = 1.                                                                                            |
| Complement                    | $P(A^c) = 1 - P(A)$ .                                                                                                 |
| Union / intersection          | $A \cup B$ : at least one; $A \cap B$ : both. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ .                             |
| Partition / total probability | If $\{B_i\}$ is mutually exclusive, exhaustive, and $P(B_i) > 0$ : $P(A) = \sum_i P(A \cap B_i)$ .                    |
| Independence                  | $A \perp B$ iff $P(A \cap B) = P(A)P(B)$ . Positive-probability mutually exclusive events are <b>not</b> independent. |
| Odds                          | $o = p/(1 - p)$ , $p = o/(1 + o)$ . Multiplying odds does not multiply probability by the same factor.                |

- “or” includes the intersection unless the question says **exactly one**.
- Equal probabilities do not imply independence.
- Frequentist probability: long-run relative frequency; Bayesian probability: belief updated by information.
- A finite-sample relative frequency estimates P(A); it is not the true probability.
- Odds micro-check: p = 0.8 gives 4:1; doubling the odds gives 8:1 and p = 8/9, not 1.6.

PMF, summaries, and transformations

For a discrete X,  $p_X(x) = P(X = x)$ ,  $p_X(x) \geq 0$ , and  $\sum_x p_X(x) = 1$ . For an event, sum the PMF over its allowed values.

- $E[X] = \sum_x x p_X(x)$ ;  $E[g(X)] = \sum_x g(x) p_X(x)$ .
- $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2$ ;  $\text{SD}(X) = \sqrt{\text{Var}(X)}$ .
- $H(X) = -\sum_x p_X(x) \log p_X(x)$ , with  $0 \log 0 := 0$ .

| Quantit y     | Key point / trap                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| Mode          | Value(s) with largest PMF; may tie; works for categories.                                                       |
| Mean          | Probability-weighted long-run average; need not be an attainable value.                                         |
| Variance / SD | $\text{Var}(X) \geq 0$ , in squared units; SD restores units. Do not confuse $E[X^2]$ with $E[X]^2$ .           |
| Entropy       | Natural log; uncertainty of the probability distribution, not magnitude of outcomes; H = 0 means no randomness. |
| Continuous X  | PDF $f_X(x)$ is a density, <b>not</b> $P(X = x)$ ; $E[g(X)] = \int g(x)f_X(x) dx$ .                             |
| Sample mean   | $\bar{X} = (1/n)\sum_i X_i$ estimates $E[X]$ ; it is not automatically the model mean.                          |

For  $Z = g(X)$ , first map every support point, then combine collisions:  $P(Z = z) = \sum_x: g(x)=z P(X = x)$ .

Example: if  $X \in \{-1, 0, 1\}$  is uniform and  $Z = X^2$ , then  $P(Z = 0) = 1/3$  and  $P(Z = 1) = 2/3$ . - If g is one-to-one on X’s support, no probabilities merge; otherwise add every collision.

Linear rules

- $E[aX + bY] = aE[X] + bE[Y]$  (no independence needed).
- $\text{Var}(aX) = a^2\text{Var}(X)$ .
- $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \text{Cov}(X, Y)$ .

Use  $E[XY] = E[X]E[Y]$  or drop the covariance term **only when X and Y are independent**.

Discrete distribution families

| Distribution / process                                          | Support and PMF                                   | Mean, variance                | R / distinction                                                                                                    |
|-----------------------------------------------------------------|---------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Bernoulli(p): one success/failure trial                         | $x \in \{0, 1\}$ ; $p^x(1 - p)^{1-x}$             | p; p(1 - p)                   | <code>dbinom(x, size = 1, prob = p)</code>                                                                         |
| Binomial(n, p): successes in n independent trials with common p | $x = 0, \dots, n$ ; $C(n, x)p^x(1 - p)^{n-x}$     | np; np(1 - p)                 | <code>dbinom(x, size = n, prob = p)</code> ; size is trials, not requested successes.                              |
| Geometric(p): failures before first success                     | $x = 0, 1, \dots$ ; $p(1 - p)^x$                  | $(1 - p)/p$ ; $(1 - p)/p^2$   | <code>dgeom(x, prob = p)</code> ; x = 0 = immediate success. “Trial number of first success” is this variable + 1. |
| NegBin(k, p): failures before k-th success                      | $x = 0, 1, \dots$ ; $C(k - 1 + x, x)p^k(1 - p)^x$ | $k(1 - p)/p$ ; $k(1 - p)/p^2$ | <code>dnbinom(x, size = k, prob = p)</code> ; size is successes. k = 1 is Geometric.                               |
| Poisson(λ): count of independent arrivals in a fixed window     | $x = 0, 1, \dots$ ; $\lambda^x e^{-\lambda} / x!$ | λ; λ                          | <code>dpois(x, lambda = lambda)</code> ; mean = variance.                                                          |

Parameters: Binomial  $0 \leq p \leq 1$  and  $n \in \{0, 1, \dots\}$ ; Geometric / NegBin  $0 < p \leq 1$  and  $k \in \{1, 2, \dots\}$ ; Poisson  $\lambda \geq 0$ . R d\* functions return  $P(X = x)$ , not a CDF or random draw.

**Selection cues:** Bernoulli = one binary trial; Binomial = successes in a fixed n trials; Geometric = failures until first success; NegBin = failures until k-th success; Poisson = arrivals counted in a fixed window. Binomial/Geometric/NegBin require independent trials with common p—do not choose only from the support.

**Family vs distribution:** a family contains all legal parameter combinations; parameters specify one distribution.

Recovering parameters from μ and v (only after family is known)

| Family    | Fast check                                                                                   |
|-----------|----------------------------------------------------------------------------------------------|
| Bernoulli | $p = \mu$ ; $v = \mu(1 - \mu)$ . Mean determines it; incompatible variance means impossible. |

| Family                 | Fast check                                                                    |
|------------------------|-------------------------------------------------------------------------------|
| Poisson                | $\lambda = \mu = v$ . Mean determines it; $\mu \neq v$ means not Poisson.     |
| Geometric              | $p = 1/(\mu + 1)$ ; $v = \mu(\mu + 1)$ . Mean determines it.                  |
| Binomial, $\mu > 0$    | $p = 1 - v/\mu$ ; $n = \mu/p$ ; require legal p and integer n.                |
| NegBin, $0 < p \leq 1$ | $p = \mu/v$ ; $k = \mu^2/(v - \mu)$ ; require $v > \mu$ and integer $k > 0$ . |

A unique legal parameter set is sufficient; multiple legal sets mean insufficient information; contradictions mean impossible. Mean and variance alone do **not** determine an arbitrary PMF.

Joint distributions, marginals, and dependence

- $p_{X,Y}(x, y) = P(X = x, Y = y)$ ;  $\sum_x \sum_y p_{X,Y}(x, y) = 1$ .
- $p_X(x) = \sum_y p_{X,Y}(x, y)$ ;  $p_Y(y) = \sum_x p_{X,Y}(x, y)$ .
- To find an event probability from a joint table, identify all allowed (x, y) cells and sum them.
- Marginals do not determine the joint distribution. Do not multiply marginal probabilities unless independence is given or verified.
- To test  $X \perp Y$ : compute both marginals, then verify  $p_{X,Y}(x, y) = p_X(x)p_Y(y)$  for **every** cell. One failure disproves independence.
- **Joint-table checks:** the whole table and each marginal PMF sum to 1; an individual row or column usually does not. Given independence and both marginals, fill every cell with  $p_X(x)p_Y(y)$ .
- $E[XY] = \sum_x \sum_y xy p_{X,Y}(x, y)$ .
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$ .
- $\rho_{XY} = \text{Cov}(X, Y) / \sqrt{\text{Var}(X)\text{Var}(Y)}$ , with  $-1 \leq \rho_{XY} \leq 1$  if both variances are positive.

| Measure / implication                                                 | What it does not establish                                                                                                        |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Independent $\Rightarrow E[XY] = E[X]E[Y]$ and $\text{Cov}(X, Y) = 0$ | —                                                                                                                                 |
| $\text{Cov}(X, Y) = 0$ or Pearson $\rho = 0$                          | Does not imply independence; it only rules out linear association.                                                                |
| Pearson $\rho$                                                        | Standardized <b>linear</b> association; undefined if either variance is 0.                                                        |
| Kendall $\tau_K$                                                      | Rank-based <b>monotonic</b> association. Without ties: $\tau_K = (\text{number concordant} - \text{number discordant})/C(n, 2)$ . |

A non-monotonic deterministic relationship such as  $Y = X^2$  can have Pearson = 0 and Kendall = 0, yet Y is fully determined by X. A strictly increasing nonlinear relationship such as  $Y = X^3$  can have  $\tau_K = 1$ .

Conditional probability and conditional independence

- For  $P(B) > 0$ :  $P(A | B) = P(A \cap B) / P(B)$ .
- Conditioning restricts the sample space to B: discard excluded outcomes, divide each retained probability by P(B), and check the resulting PMF sums to 1.
- Micro-example:  $p_X(1, 2, 3, 4) = (0.1, 0.2, 0.3, 0.4)$ . Given  $X \geq 3$ ,  $P(X \geq 3) = 0.7$ , so  $P(X = 3 | X \geq 3) = 3/7$  and  $P(X = 4 | X \geq 3) = 4/7$ ;  $X = 1, 2$  get probability 0.
- $P(Y = y | X = x) = P(Y = y, X = x) / P(X = x)$ , for  $P(X = x) > 0$ .
- $P(Y = y, X = x) = P(Y = y | X = x)P(X = x)$ .
- $P(Y = y) = \sum_x P(Y = y | X = x)P(X = x)$ .
- $E[Y] = \sum_x E[Y | X = x]P(X = x) = E_X[E[Y | X]]$ .
- $P(A | B) \neq P(B | A)$  in general: denominators differ.
- Under  $X \perp Y$ ,  $P(Y = y | X = x) = P(Y = y)$ : learning X does not change Y’s distribution.
- For total expectation, weight conditional means by  $P(X = x)$ ; do not take an unweighted average.

$X \perp Y | Z$  iff  $P(X = x, Y = y | Z = z) = P(X = x | Z = z)P(Y = y | Z = z)$  for every z with  $P(Z = z) > 0$ .

To test conditional independence: for each z, restrict the three-way joint distribution to  $Z = z$ , renormalize, obtain conditional marginals, and test factorization across all (x, y). Marginal independence and conditional independence do not imply one another; use the distributional test, not a correlation coefficient.

**Binary shortcut:** for fixed z with known conditional marginals, checking  $P(X = 1, Y = 1 | Z = z) = P(X = 1 | Z = z)P(Y = 1 | Z = z)$  determines whether the other three cells factorize; with more categories, test every cell.